

JOSHITHA RAJASEKARAN

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SUMMARY

Computer Science Engineering student (CGPA 8.3/10) with strong hands-on expertise in Python, Machine Learning, and AWS Cloud Computing. Built and deployed scalable AI-powered applications — a Speech-to-Text system with 18% accuracy improvement and a real-time Accident Detection system with 92% precision. Proven ability to manage cloud infrastructure, solve complex problems, and collaborate effectively in fast-paced environments.

CORE SKILLS

Languages	Python 3.x (proficient), Java (fundamentals), C
Python Libraries	TensorFlow 2.x, Flask, Librosa, SciPy, NumPy, Pandas
Cloud (AWS)	EC2, S3, IAM, CloudWatch
Web & Tools	HTML5, Basic CSS3, Git, GitHub
Soft Skills	Problem Solving, Analytical Thinking, Team Collaboration, Adaptability, Communication

EXPERIENCE

Cloud Computing Intern — Orionshift

Jul 2025 – Aug 2025

- Managed 5+ AWS EC2 instances and S3 buckets supporting 2 production application deployments.
- Reduced infrastructure setup time by ~30% through documented configuration runbooks.
- Resolved 10+ infrastructure incidents, maintaining 99% service uptime.
- Enforced IAM security policies and monitored resources via CloudWatch dashboards.
- Collaborated with team members to ensure smooth cloud operations and deployment activities.

PROJECTS

Speech-to-Text Transcription System

Python, TensorFlow 2.x, Flask, Librosa, SciPy

- Trained a deep learning model achieving 18% accuracy improvement using noise augmentation on 500+ audio samples.
- Extracted MFCC/spectrogram features with Librosa and SciPy, cutting preprocessing time by 40%.
- Deployed as a Flask web application for real-time audio upload and transcription.
- Built an end-to-end pipeline covering data processing, model training, and web-based interaction.

Smart Accident Detection and Emergency Human Injury Alert System

Python, TensorFlow, OpenCV, NumPy

- Developed a real-time accident detection system using computer vision and deep learning to identify vehicle collisions and human injuries from surveillance feeds.
- Implemented automated emergency alert mechanism to notify the nearest hospital and emergency contacts with GPS-based incident location within seconds of detection.
- Achieved 92% accident detection accuracy by training a CNN model on 1,000+ labeled accident and non-accident image datasets with data augmentation techniques.
- Worked on real-time monitoring, data processing, and automated response mechanisms.

EDUCATION

B.E. Computer Science and Engineering

University College of Engineering, Thirukkuvalai | 2022 – 2026 | CGPA: 8.3 / 10.0

Relevant Coursework: Data Structures & Algorithms, OS, Computer Networks, DBMS, Machine Learning, Cloud Computing

CERTIFICATIONS & ACHIEVEMENTS

- Cloud Computing — Orionshift (2025)
- Full Stack Web Development — Skill Intern (2024)

- Academic Rank Holder — Computer Science Engineering, University College of Engineering